

Quiz 3

1. ROC and DTFT

$X(z) = 1/(1 - 1.25z^{-1})$ with ROC $|z| < 1.25$. Does the DTFT exist? Is the corresponding sequence right-sided or left-sided?

2. Inverse z-transform

Find $x[n]$ for $X(z) = 1/(1 - 0.6z^{-1})$, ROC $|z| > 0.6$.

3. Transfer function

For $y[n] - 0.5y[n - 1] = x[n] - x[n - 1]$, find $H(z)$, its pole and its zero.

4. Causal and stable rule

A causal rational system has a pole at $z=1.02$. Can it be BIBO stable? Briefly justify.

5. Geometry

What happens to $|H(e^{j\omega})|$ at a frequency whose unit-circle point coincides exactly with a zero?

What happens qualitatively near a pole that lies very close to the unit circle?